cogs 9 | Week 4

Data visualization

Brendan Tomoschuk

Exam 1 Summary

- **Stats:** Mean = 31.8/35 = 90.8%
- This is very good!
- …but midterm 2 is usually harder (Monday Week 9)
- If you didn't do as well as you hoped, *we want to help!*
  - Come see us in section / visit us in office hours
- Individual exam scores will be released Friday
- **If you would like to review your exam:**
  - Wednesdays, 11-12pm
  - book a 1-1 with me Thursdays 1-2

Guest Lecture — Friday, October 23, 2026

Data Exploration & Startup life

- Participation is worth **1 point**
  - Fill out a Google Form of your takeaways
  - Interactive **Q&A!**
- Dewey data organizes and sells datasets to academic researchers
- Questions examples:
  - Working at at a small company?
  - What kind of data do you work with?

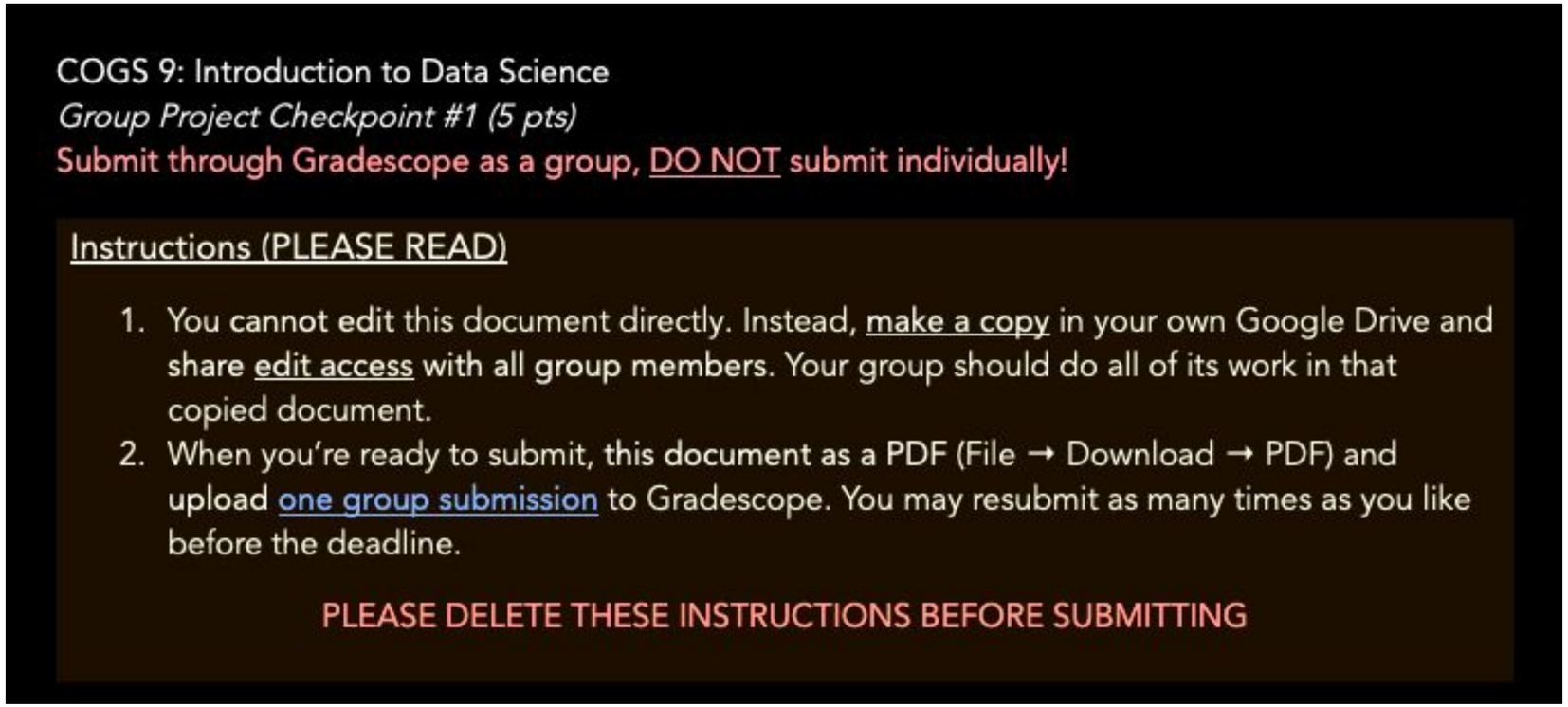


Taylor Butler

Research & Growth Lead

Group Project - Checkpoint 1 due Wed week 5

- You should all have your groups assigned
- Remember that if you don't respond do your group in a timely manner they have permission to drop you!
- Checkpoint 1



COGS 9: Introduction to Data Science

Group Project Checkpoint #1 (5 pts)

Submit through Gradescope as a group, DO NOT submit individually!

Instructions (PLEASE READ)

1. You cannot edit this document directly. Instead, make a copy in your own Google Drive and share edit access with all group members. Your group should do all of its work in that copied document.
2. When you're ready to submit, this document as a PDF (File → Download → PDF) and upload one group submission to Gradescope. You may resubmit as many times as you like before the deadline.

PLEASE DELETE THESE INSTRUCTIONS BEFORE SUBMITTING

Reading(s) 4: *Data Visualization*

Main idea: visualization design choices should be tailored to your audience

Data is Personal: Attitudes and Perceptions of Data Visualization in Rural Pennsylvania

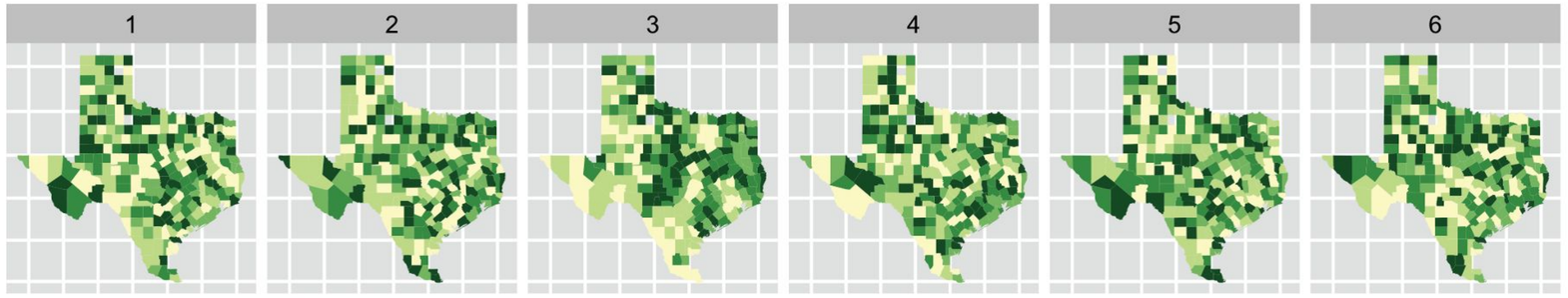
Evan M. Peck
Bucknell University
evan.peck@bucknell.edu

Sofia E. Ayuso
Bucknell University
sea018@bucknell.edu

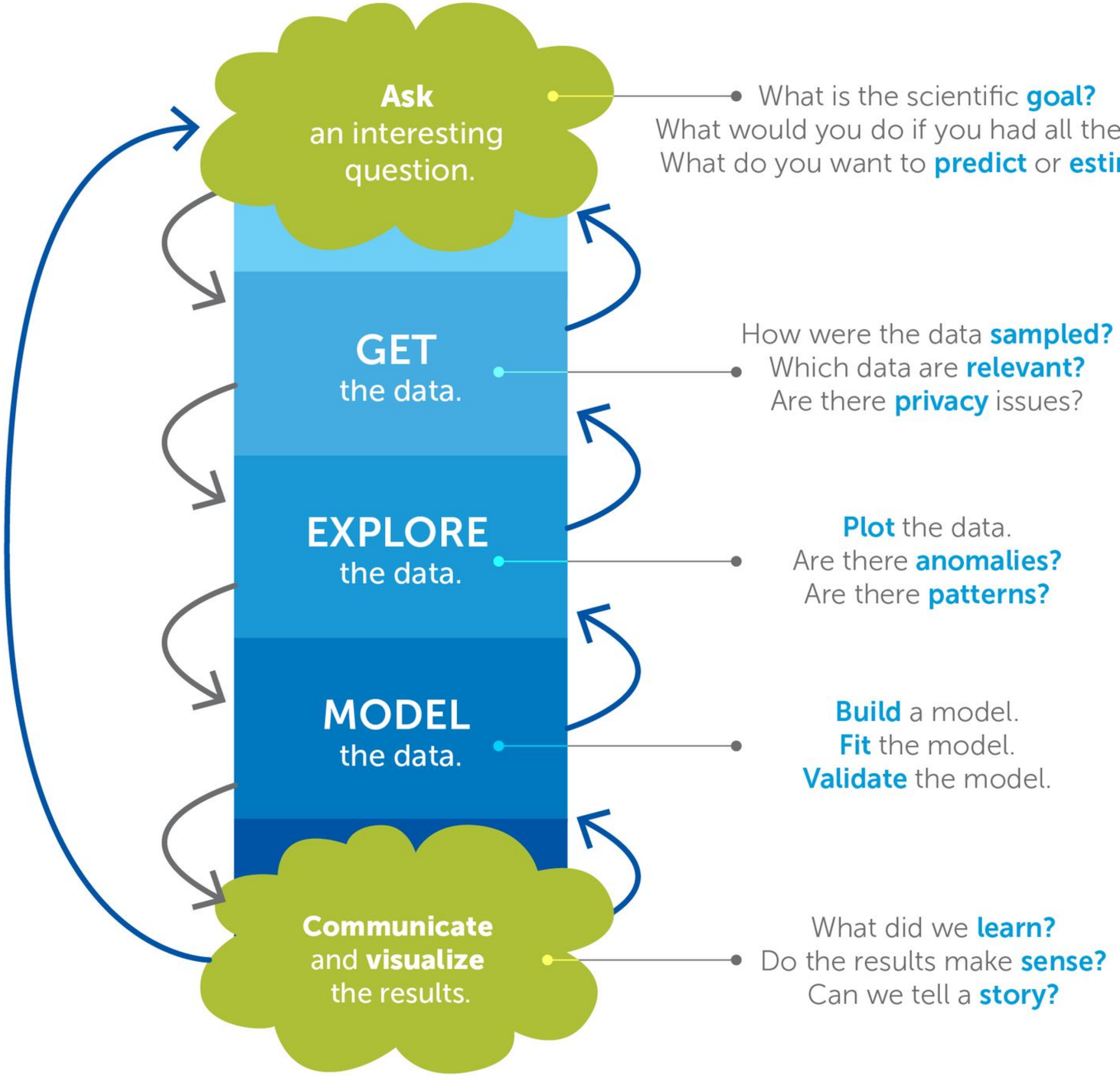
Omar El-Etr
Bucknell University
omar.eletr@bucknell.edu

Graphical Inference for Infovis

Hadley Wickham, Dianne Cook, Heike Hofmann, and Andreas Buja



Main idea: Visualizations are powerful tools for understanding structure in data



So far:

- What is data science
- Asking good questions
- Ethics & privacy
- Data types and storage
- Getting & cleaning data

Up next:

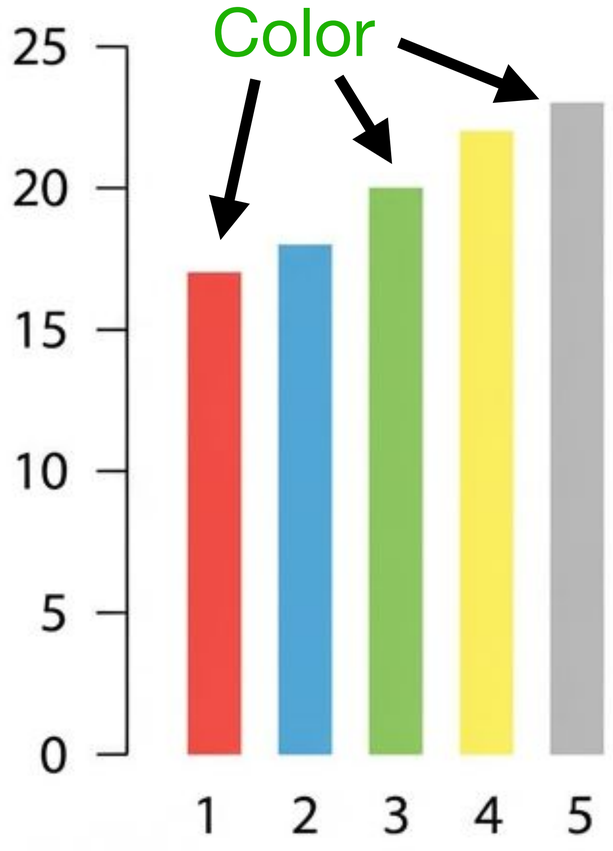
- Visualizing & exploring data
- Modeling the data (ML)
- Algorithmic bias & fairness
- Data science applications
- Future of data science

Visual encodings

Quantitative encodings: *magnitude*

From **most to least** precise:

1. **Position**
2. **Length**
3. **Angle**
4. **Area**
5. **Color**



Bar plot

Length of bars encodes the values

Position on x-axis indicates different categories

Qualitative encodings: *categories*

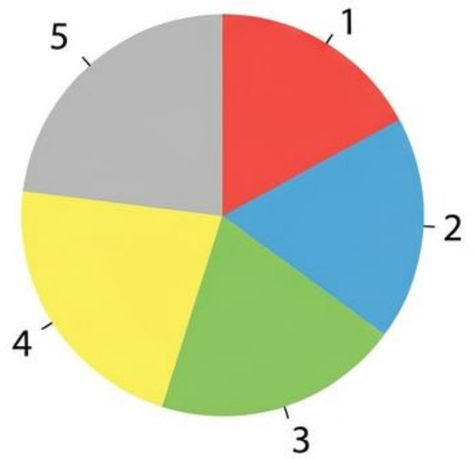
1. **Position**
2. **Color**
3. **Shape**
4. **Texture/pattern**

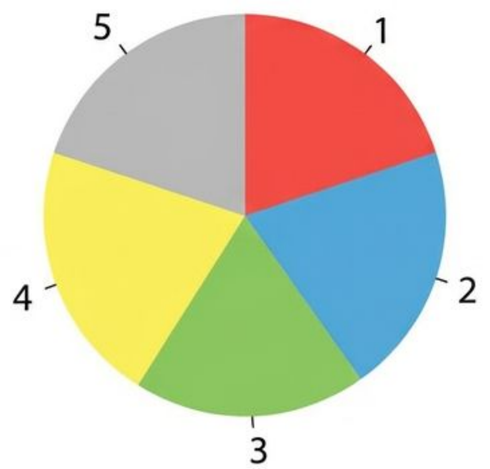
Precision of encodings

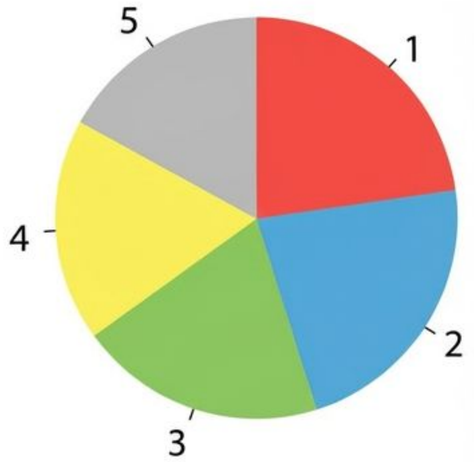
Which category is the largest?

Encoding: Area

Pie charts are *not precise enough* to compare values across categories

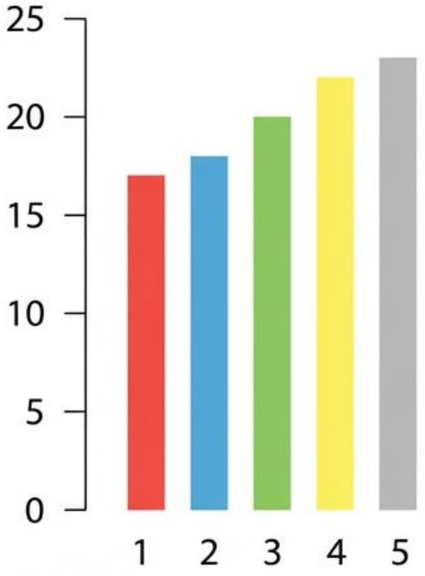


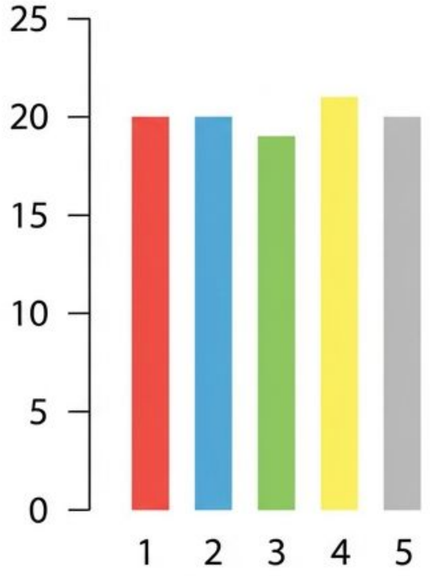


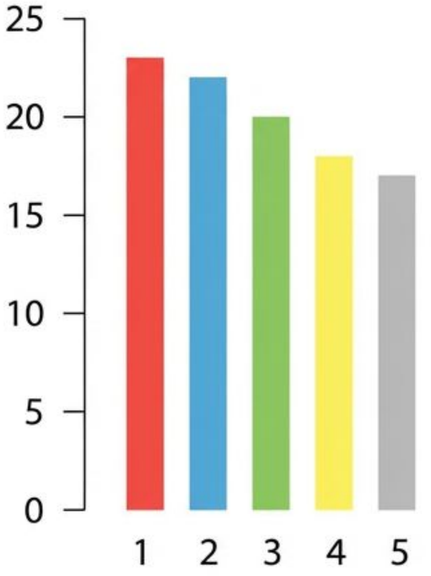


Encoding: Length

Bar charts are a more precise way to compare values across categories







Webclicker

Sign in with your UCSD email!

webclicker.web.app



course code:

WSTAPX

You want to know the exact percentage of each major represented in COGS 9. Which of these visualizations is the best choice?

(A) Pie chart

(B) Bar chart

(C) Just the raw data table

(D) Line plot

(E) I'm not sure / I'm confused

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You want to know the exact percentage of each major represented in COGS 9. Which of these visualizations is the best choice?

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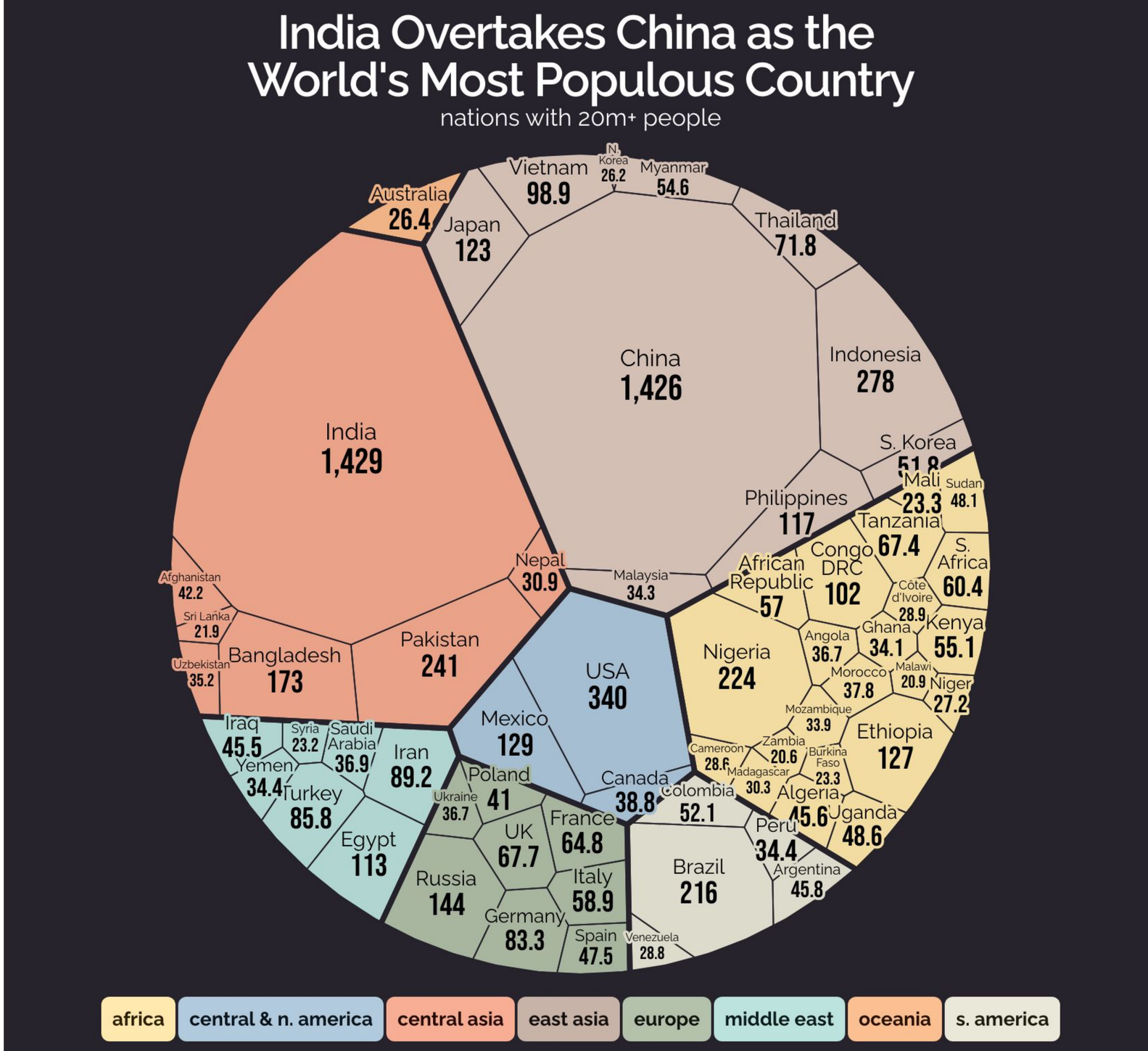
What makes a visualization effective?

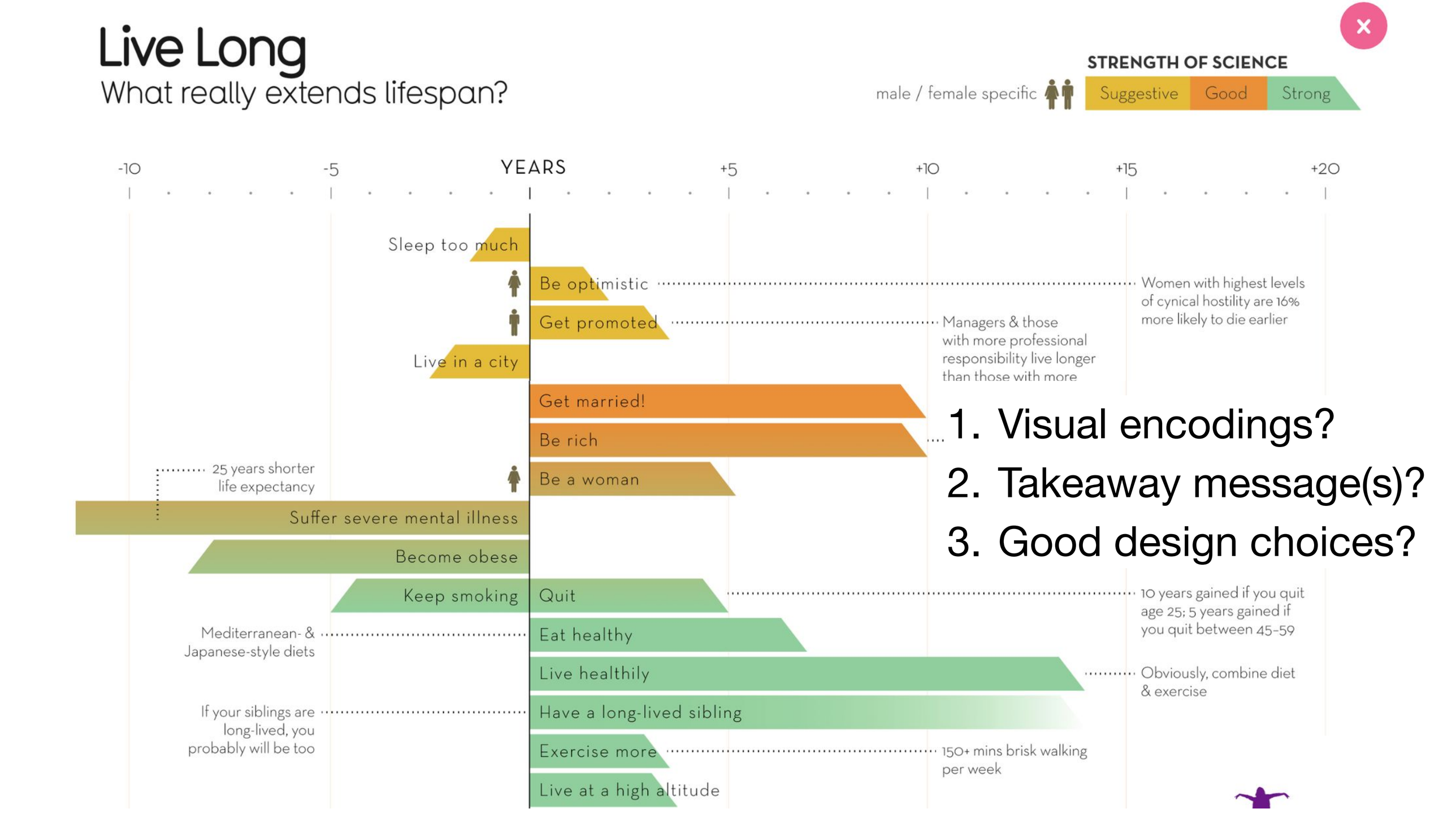
(~2 min with your neighbor)

For each of the following visualizations…

1. What visual encodings are used?
2. Takeaway message(s) from this plot?
3. What design choices help you understand the plot easily?

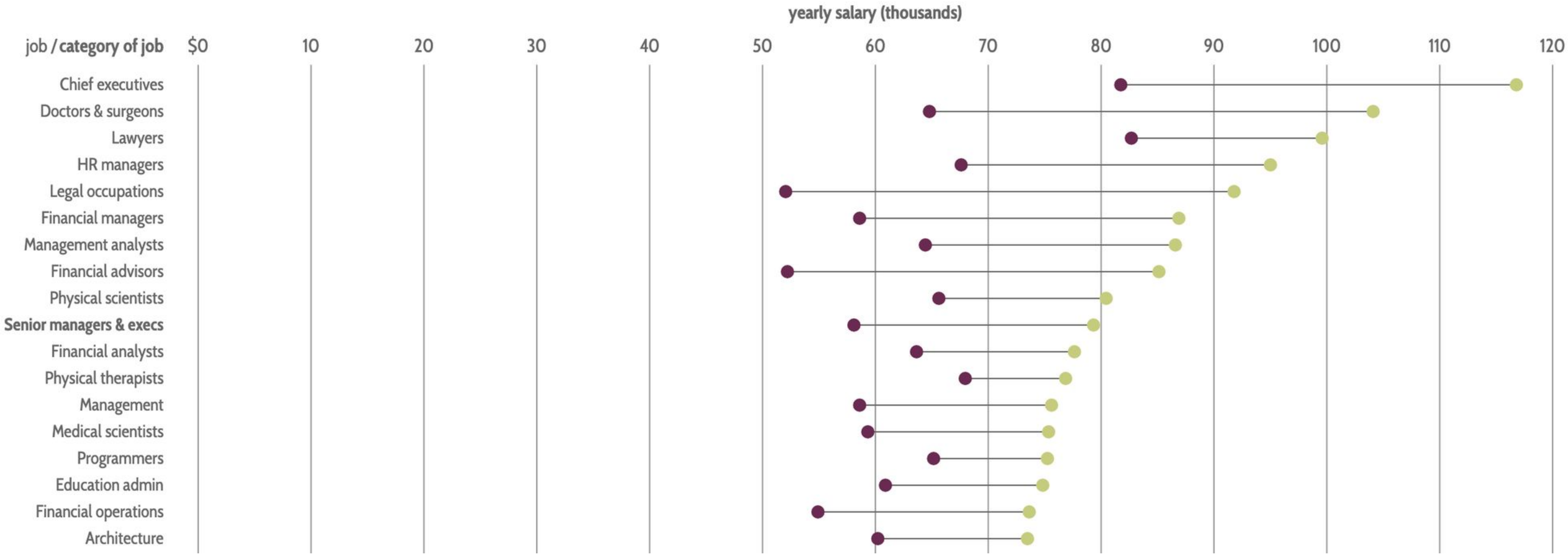
1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?





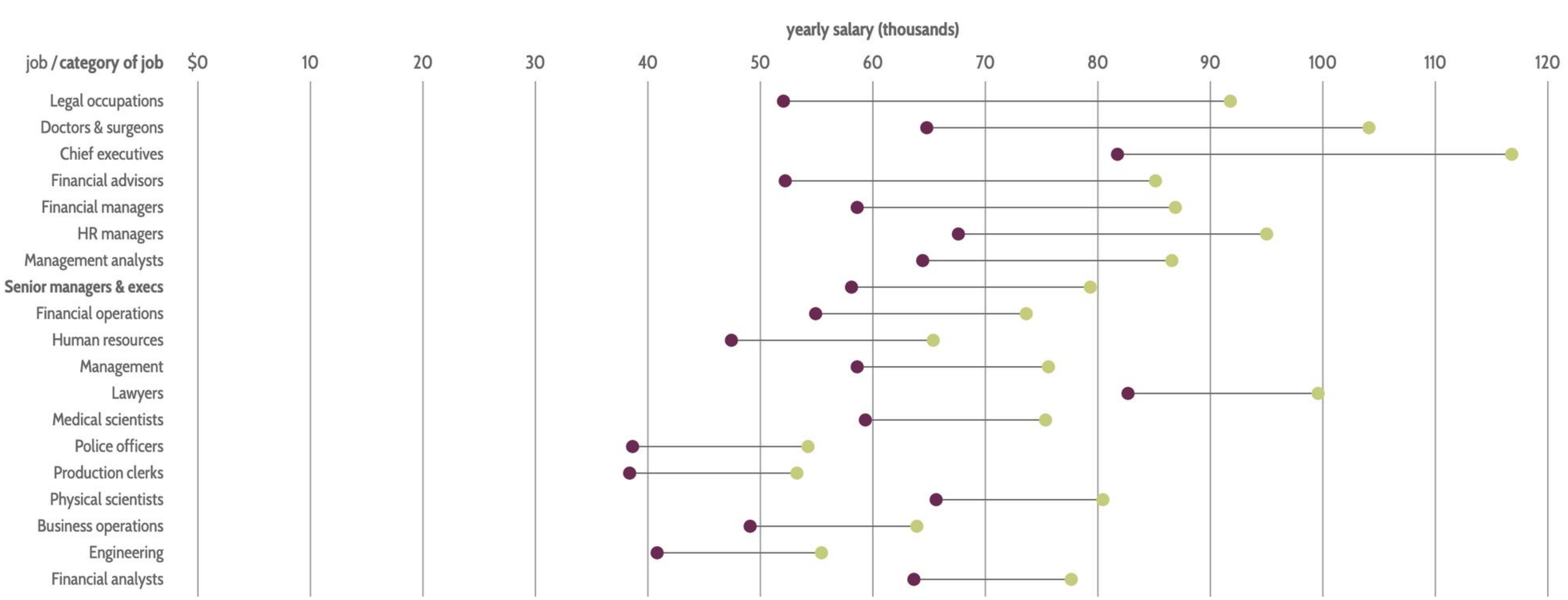
1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?

The Gender Pay Gap



1. Visual encodings?
2. Takeaway message(s)?
3. Good design choices?

The Gender Pay Gap



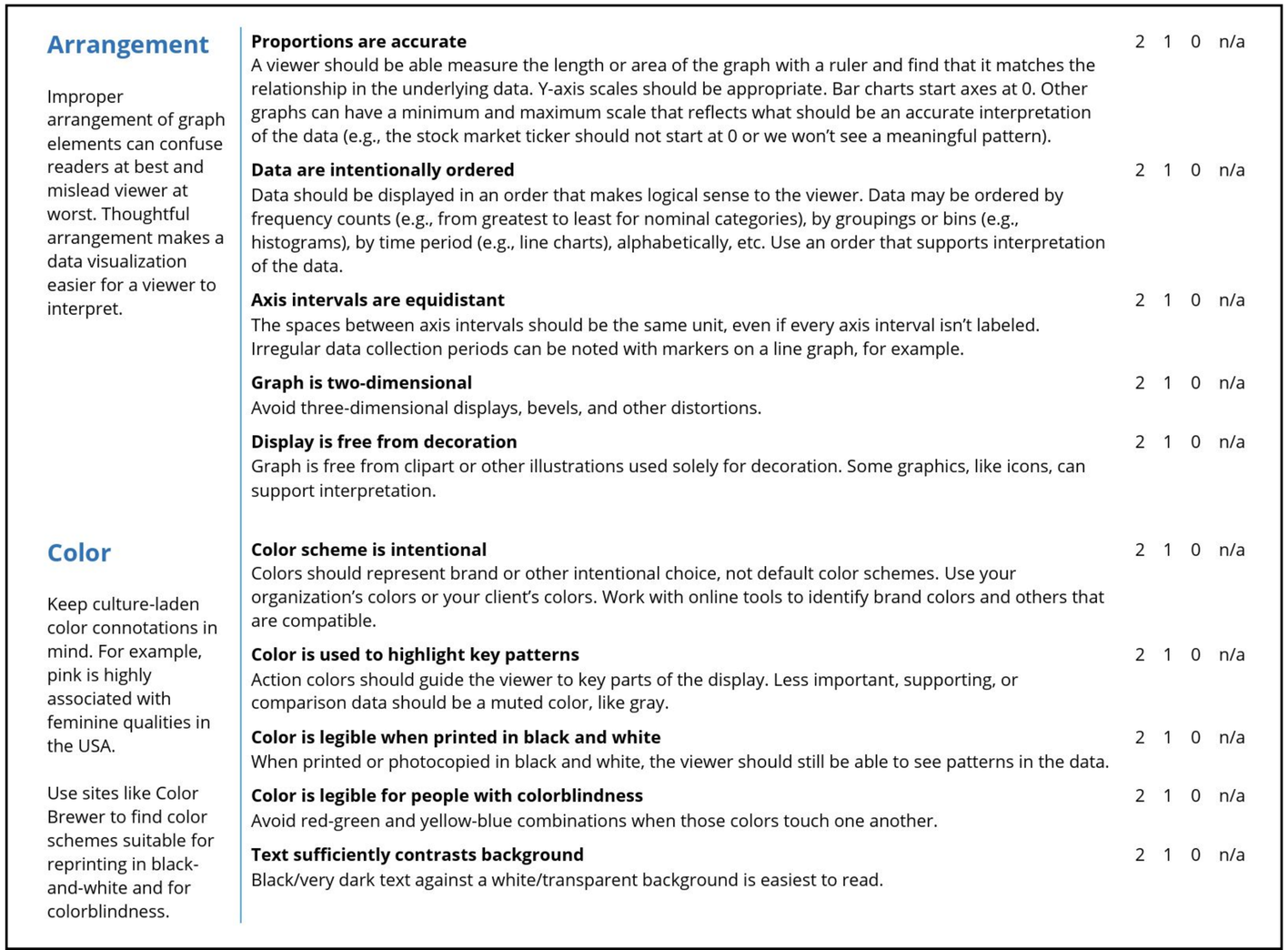
What does this version of the plot highlight that the other version did not?

Data Visualization Checklist

24 items regarding a visualization's

datavisualizationchecklist.com

1. Text/labels
2. Arrangement
3. Color choices
4. Lines and axes
5. Overall plot choice and message



Arrangement

Improper arrangement of graph elements can confuse readers at best and mislead viewer at worst. Thoughtful arrangement makes a data visualization easier for a viewer to interpret.

Proportions are accurate 2 1 0 n/a
A viewer should be able measure the length or area of the graph with a ruler and find that it matches the relationship in the underlying data. Y-axis scales should be appropriate. Bar charts start axes at 0. Other graphs can have a minimum and maximum scale that reflects what should be an accurate interpretation of the data (e.g., the stock market ticker should not start at 0 or we won't see a meaningful pattern).

Data are intentionally ordered 2 1 0 n/a
Data should be displayed in an order that makes logical sense to the viewer. Data may be ordered by frequency counts (e.g., from greatest to least for nominal categories), by groupings or bins (e.g., histograms), by time period (e.g., line charts), alphabetically, etc. Use an order that supports interpretation of the data.

Axis intervals are equidistant 2 1 0 n/a
The spaces between axis intervals should be the same unit, even if every axis interval isn't labeled. Irregular data collection periods can be noted with markers on a line graph, for example.

Graph is two-dimensional 2 1 0 n/a
Avoid three-dimensional displays, bevels, and other distortions.

Display is free from decoration 2 1 0 n/a
Graph is free from clipart or other illustrations used solely for decoration. Some graphics, like icons, can support interpretation.

Color

Keep culture-laden color connotations in mind. For example, pink is highly associated with feminine qualities in the USA.

Use sites like Color Brewer to find color schemes suitable for reprinting in black-and-white and for colorblindness.

Color scheme is intentional 2 1 0 n/a
Colors should represent brand or other intentional choice, not default color schemes. Use your organization's colors or your client's colors. Work with online tools to identify brand colors and others that are compatible.

Color is used to highlight key patterns 2 1 0 n/a
Action colors should guide the viewer to key parts of the display. Less important, supporting, or comparison data should be a muted color, like gray.

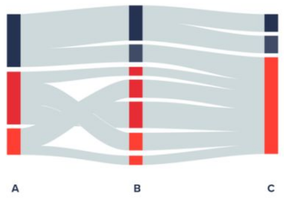
Color is legible when printed in black and white 2 1 0 n/a
When printed or photocopied in black and white, the viewer should still be able to see patterns in the data.

Color is legible for people with colorblindness 2 1 0 n/a
Avoid red-green and yellow-blue combinations when those colors touch one another.

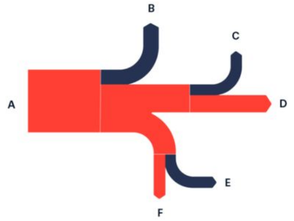
Text sufficiently contrasts background 2 1 0 n/a
Black/very dark text against a white/transparent background is easiest to read.

How do I choose the right visualization for my data?

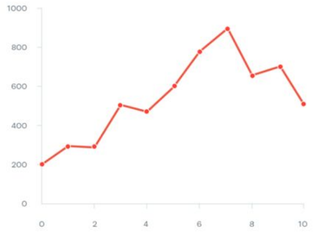
Alluvial Diagram



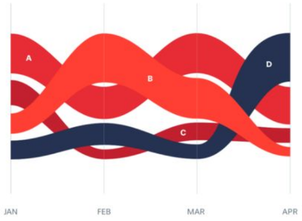
Sankey Diagram



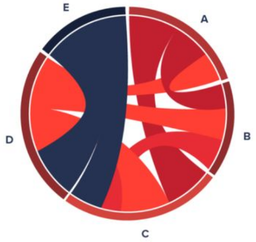
Line Graph



Sorted Stream Graph



Chord Diagram



Proportional Area Chart (Square)

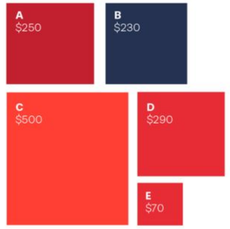
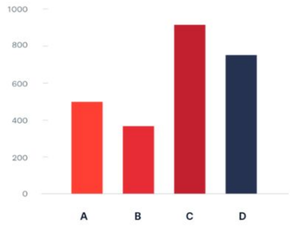


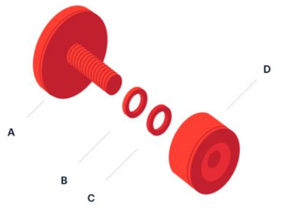
Table Chart

	A	B	C
X	$40	240	48
Y	$50	200	59
Z	$60	310	79

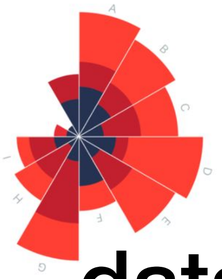
Bar Chart (Vertical)



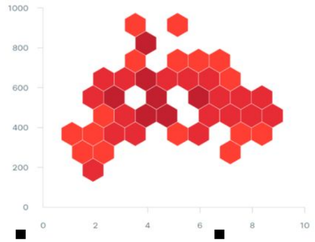
Exploded View Drawing



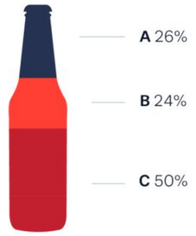
Polar Area Chart



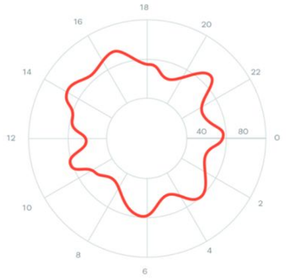
Hexagonal Binning



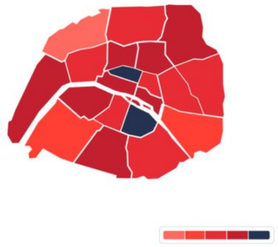
Pictorial Stacked Chart



Radial Line Graph



Choropleth Map

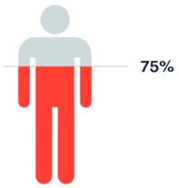


datavizproject.com

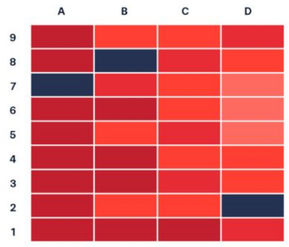
Sunburst Diagram



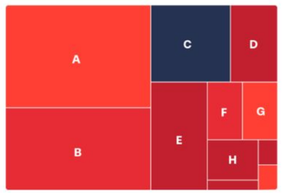
Pictorial Percentage Chart



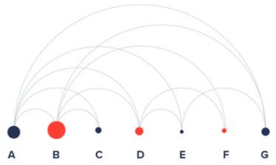
Heat Map



Treemap



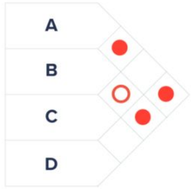
Arc Diagram



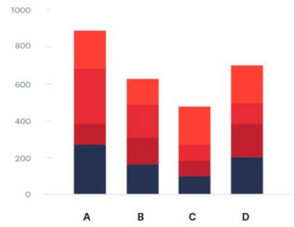
Progress Bar



Matrix Diagram (Roof Shaped)



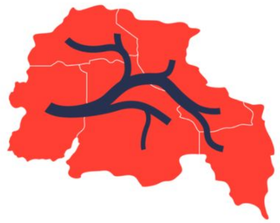
Stacked Bar Chart



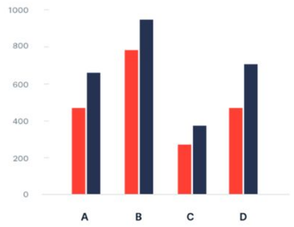
Radial Histogram



Flow Map



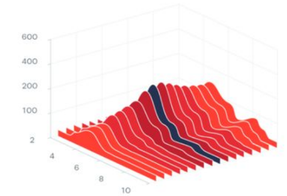
Grouped Bar Chart



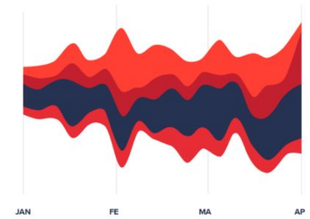
Pictorial Fraction Chart



Waterfall Plot



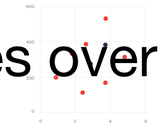
Stream Graph



The visualization should fit the *data & question*

Two questions to ask yourself before choosing a visualization:

1. How many variables do I have?
   - 1 variable → *distribution* (histogram, box plot)
   - 2 variables → *relationship or comparison* (scatter, bar, line)
2. What am I trying to show? What point am I trying to make?
   - *Comparison of values across groups* → bar plot
   - *Trend over time* → line plot
   - *Percentage of a whole* → pie chart

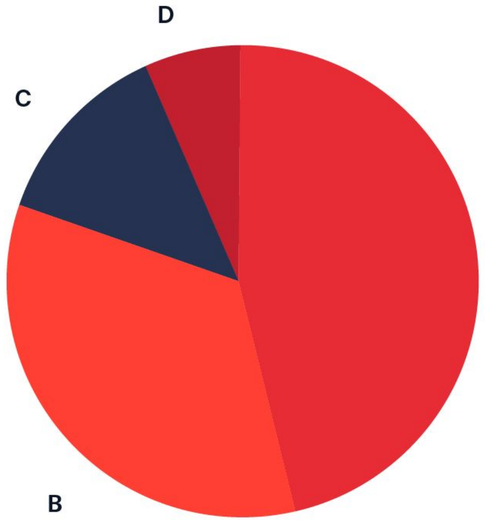
Visualization	Data Type	# of Variables	Purpose
Box plot	Quantitative	1	Summarize distribution (median, quartiles, outliers)
Histogram	Quantitative	1	Show count distribution (shape, spread, skew) of a variable
Pie chart	Qualitative OR Quantitative	1	Show proportions (few categories only)
Line chart	Quantitative	2	Show trends or changes over time 
Bar chart	Qualitative + Quantitative	2	Compare values across categories
Heatmap	Quantitative	2	Show patterns/intensity across two dimensions
Scatter plot	Quantitative	2	Show relationship/correlation between two variables

Webclicker

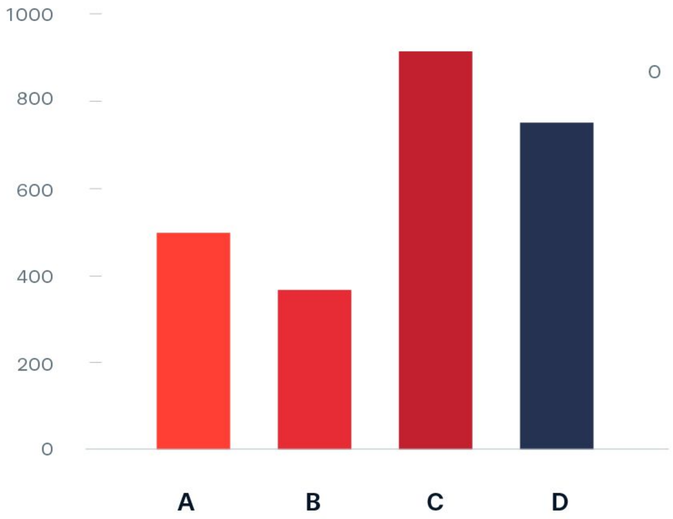
webclicker.web.app

I want to know the distribution of Instagram followers of students in COGS 9. Which visualization is best suited for this?

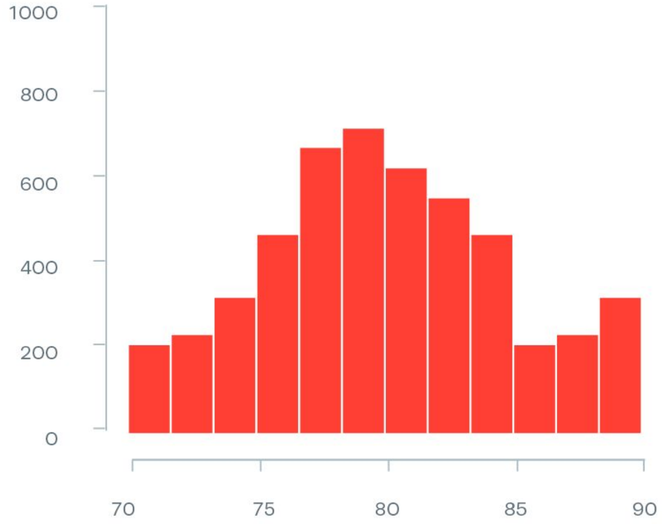
(A) pie chart



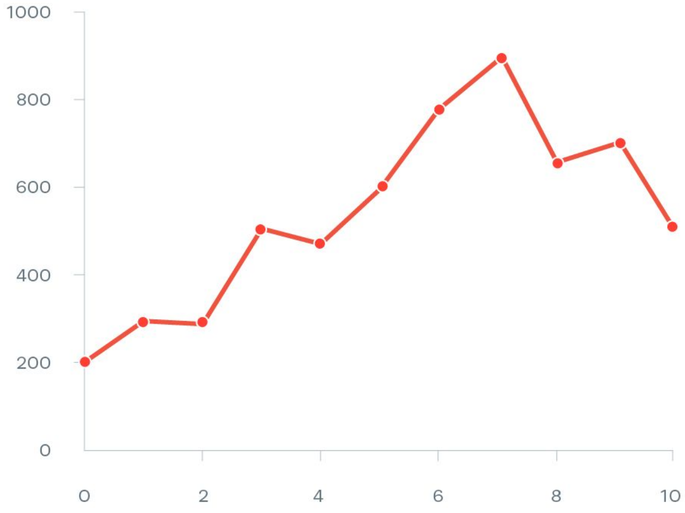
(B) bar chart



(C) histogram



(D) line plot



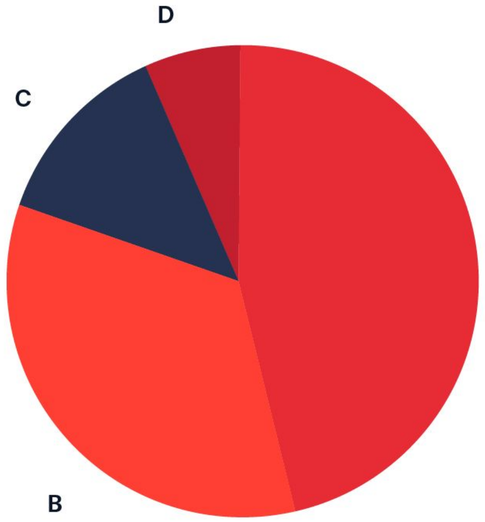
(E) I'm not sure / I'm confused

Webclicker

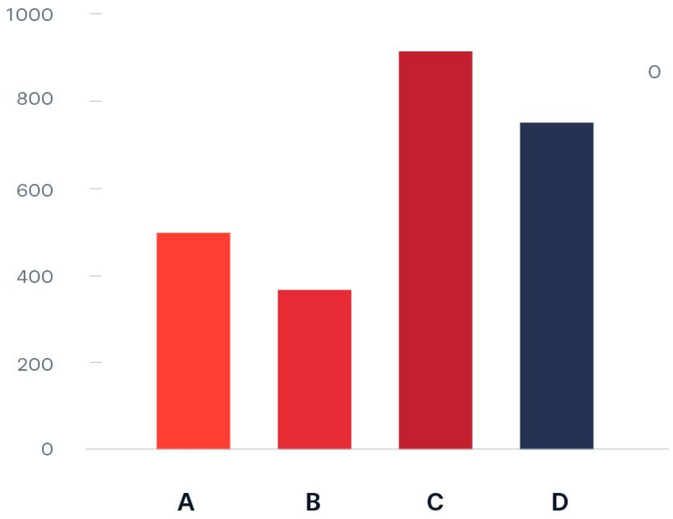
webclicker.web.app

I want to know the distribution of Instagram followers of students in COGS 9. Which visualization is best suited for this?

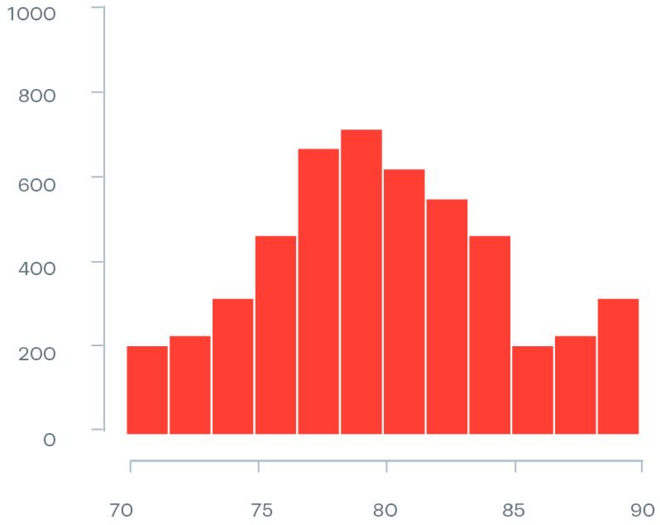
(A) pie chart



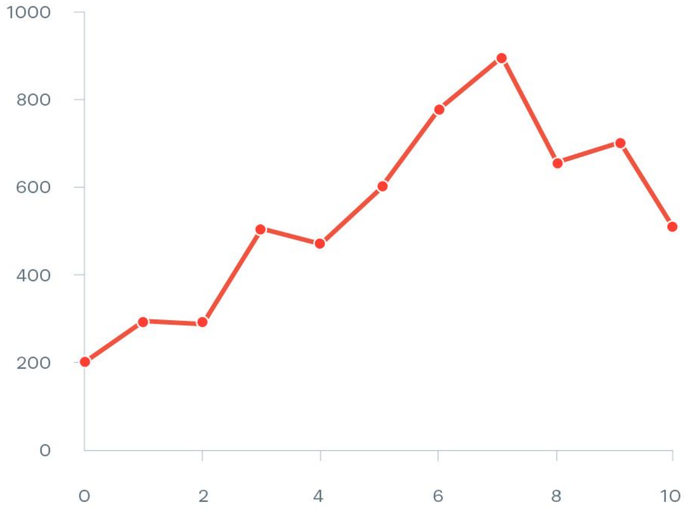
(B) bar chart



(C) histogram



(D) line plot



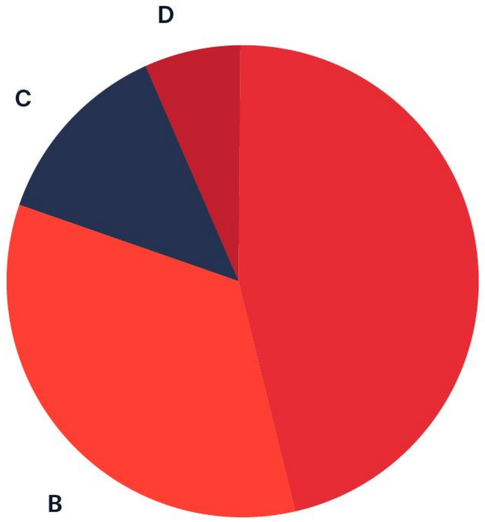
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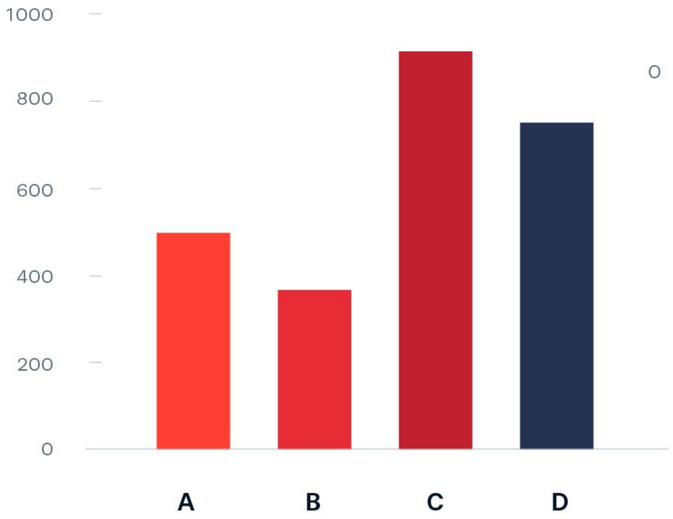
webclicker.web.app

I want to compare the average # of Instagram followers of 1st-year vs. 2nd-year students in COGS 9. Which visualization is best suited for this?

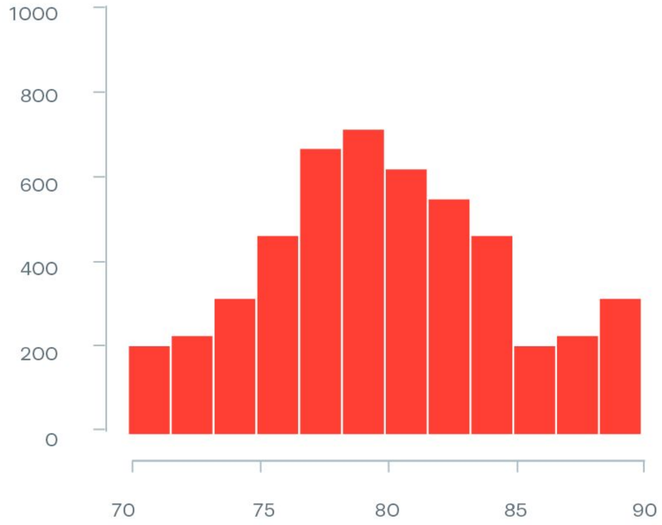
(A) pie chart



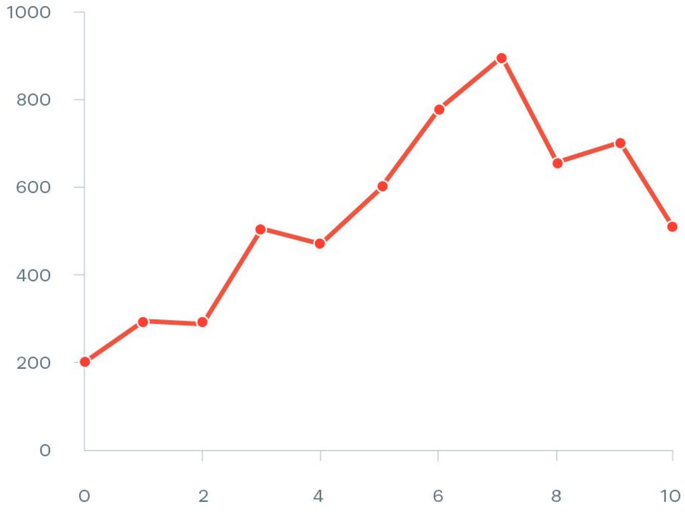
(B) bar chart



(C) histogram



(D) line plot



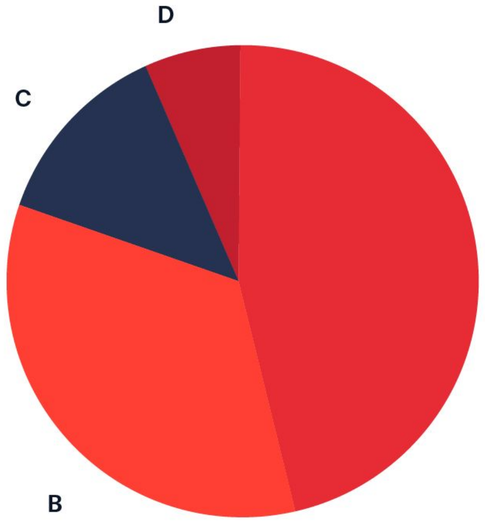
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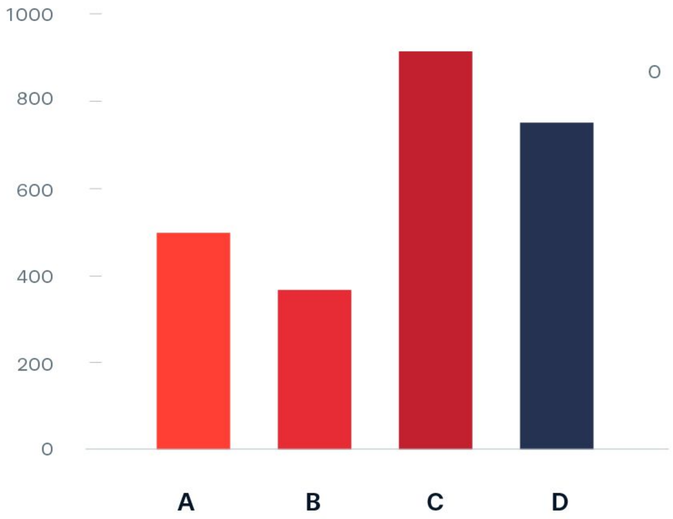
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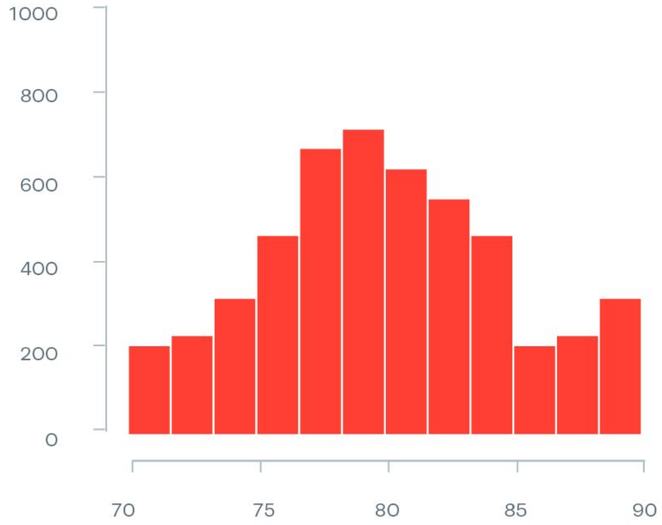
(A) pie chart



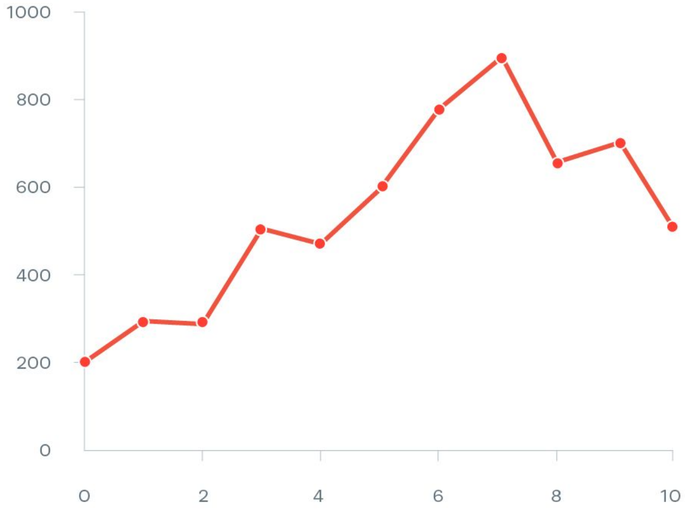
(B) bar chart



(C) histogram



(D) line plot



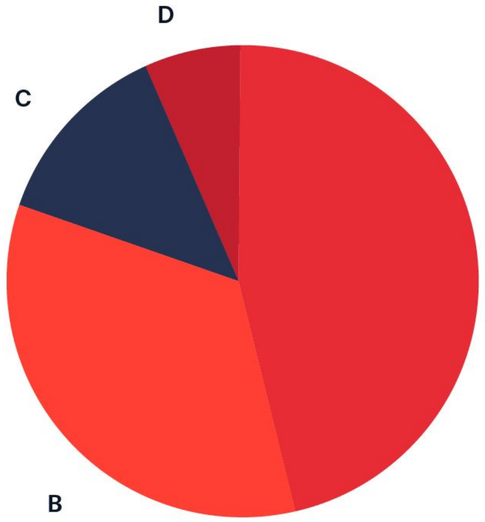
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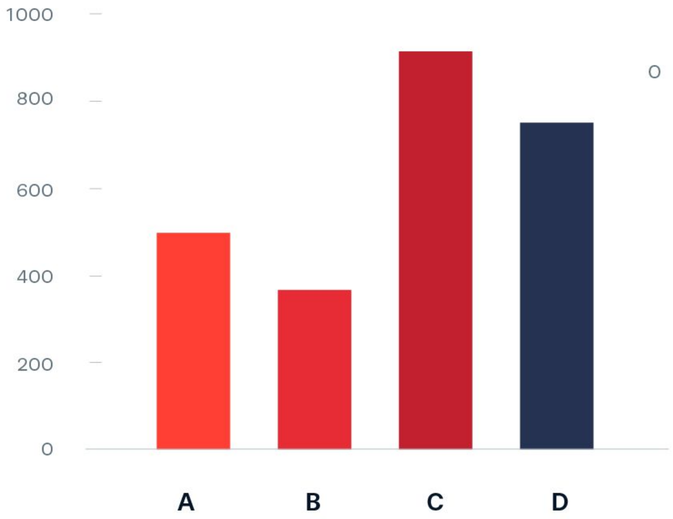
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I want to understand how many students are coming to COGS 9 lecture over the course of this quarter. Which visualization is best suited for this?

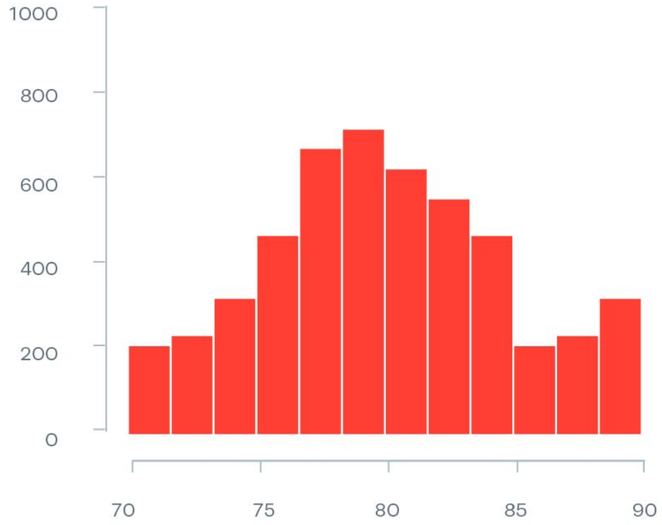
(A) pie chart



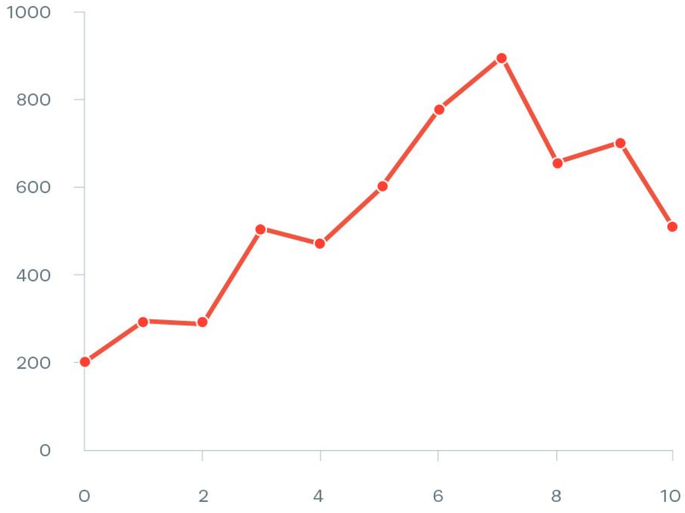
(B) bar chart



(C) histogram



(D) line plot



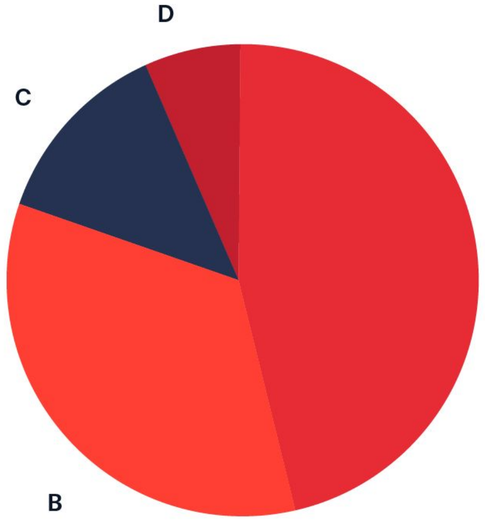
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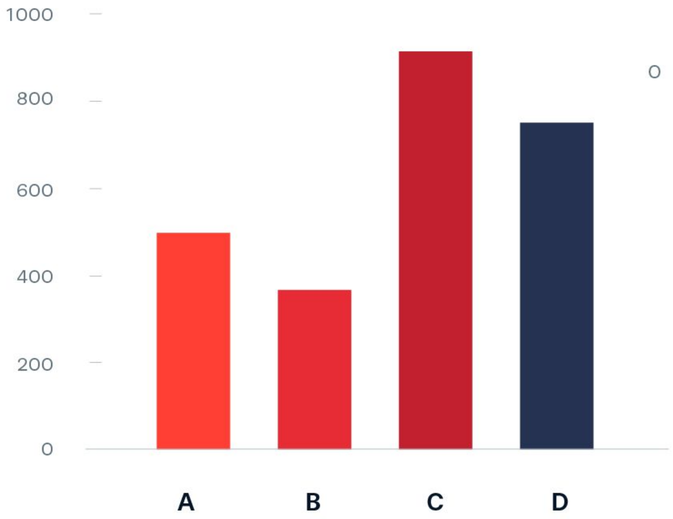
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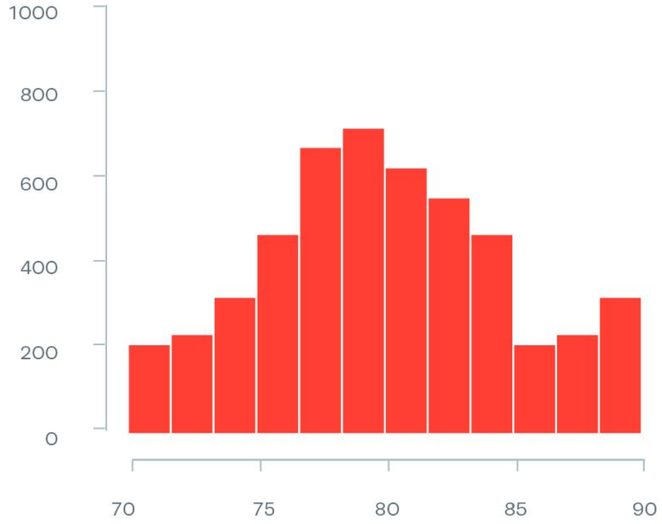
(A) pie chart



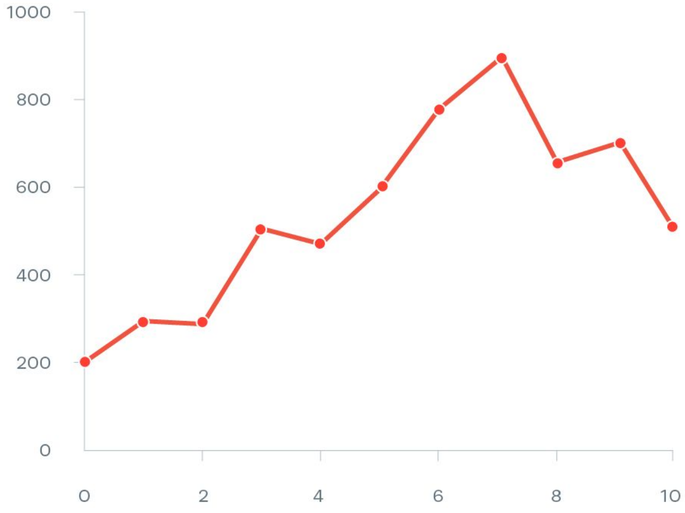
(B) bar chart



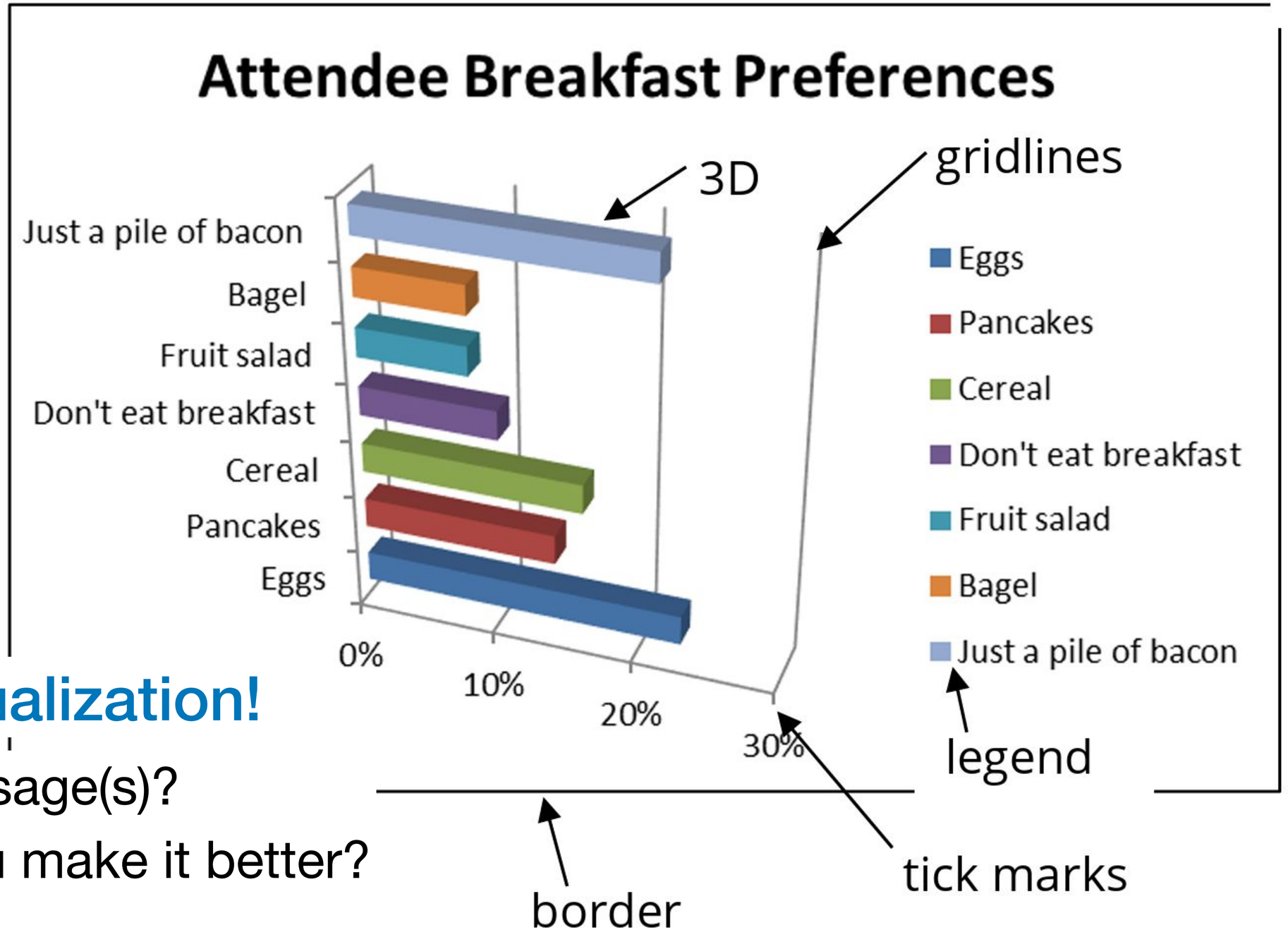
(C) histogram



(D) line plot

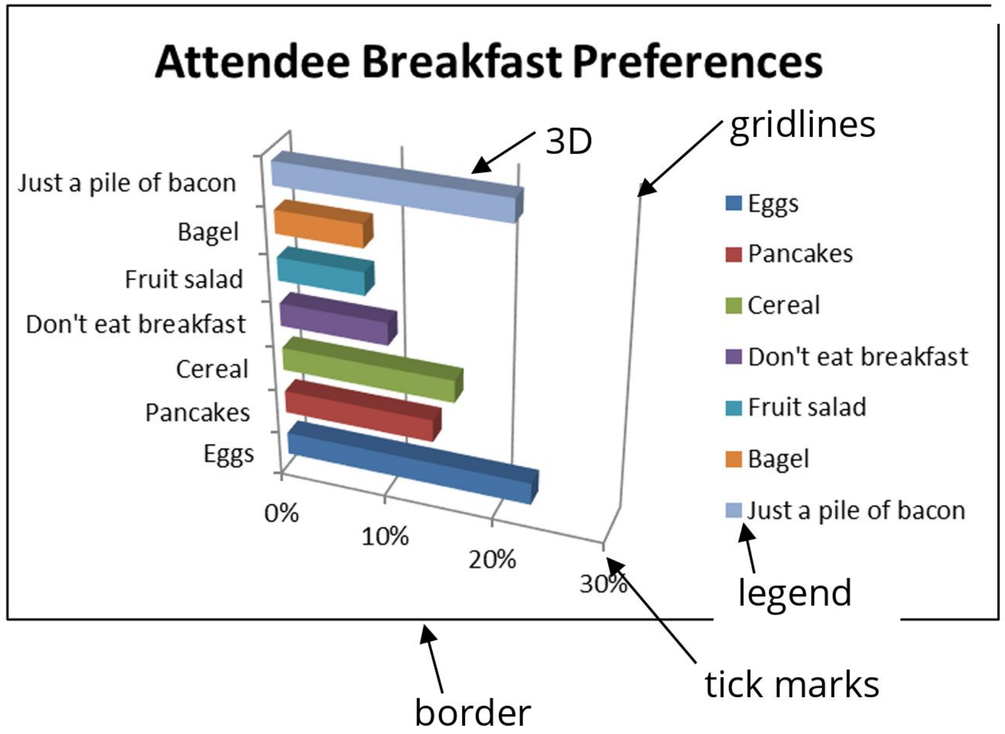


(E) I'm not sure / I'm confused



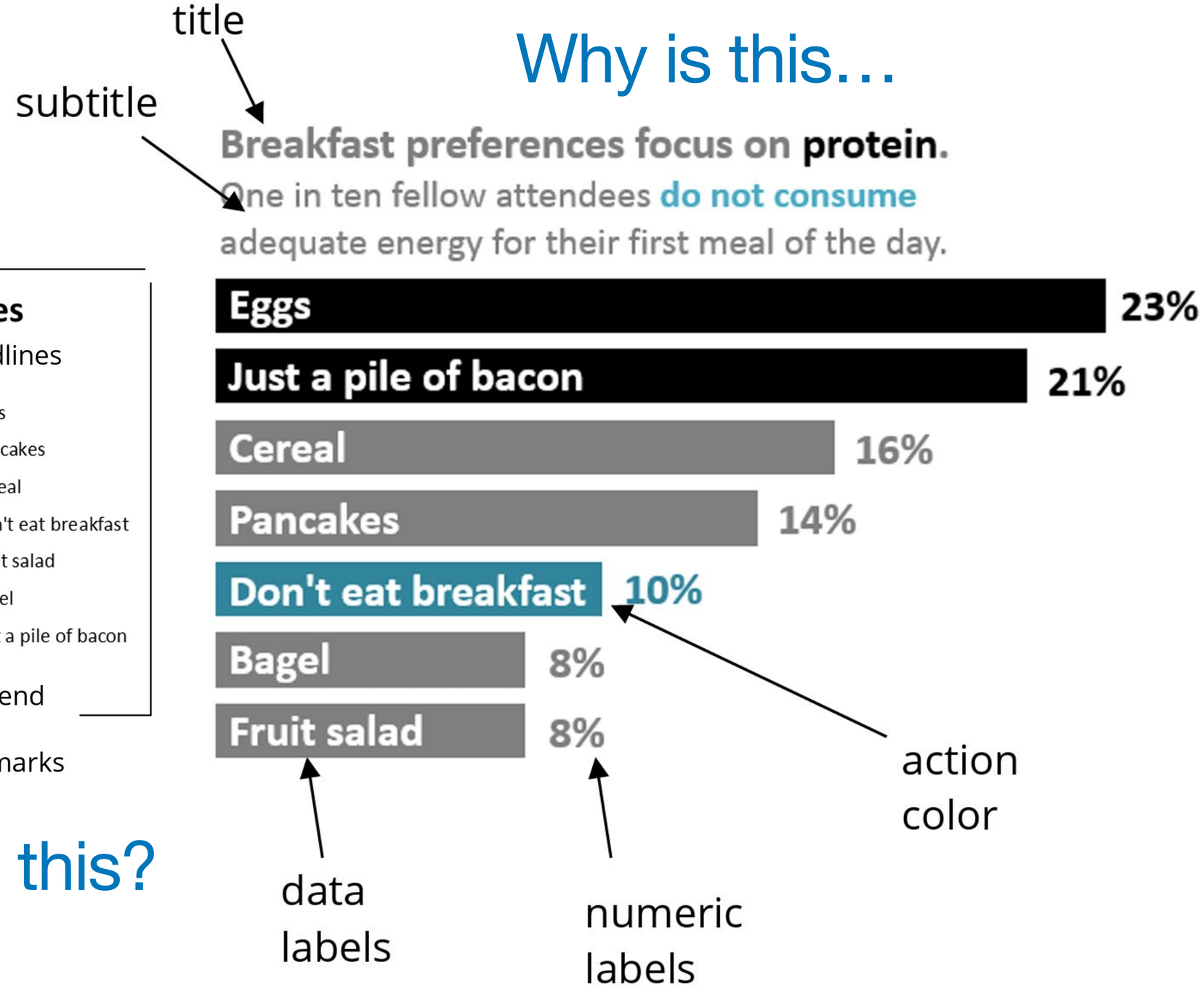
Judge this visualization!

1. Takeaway message(s)?
2. How would you make it better?



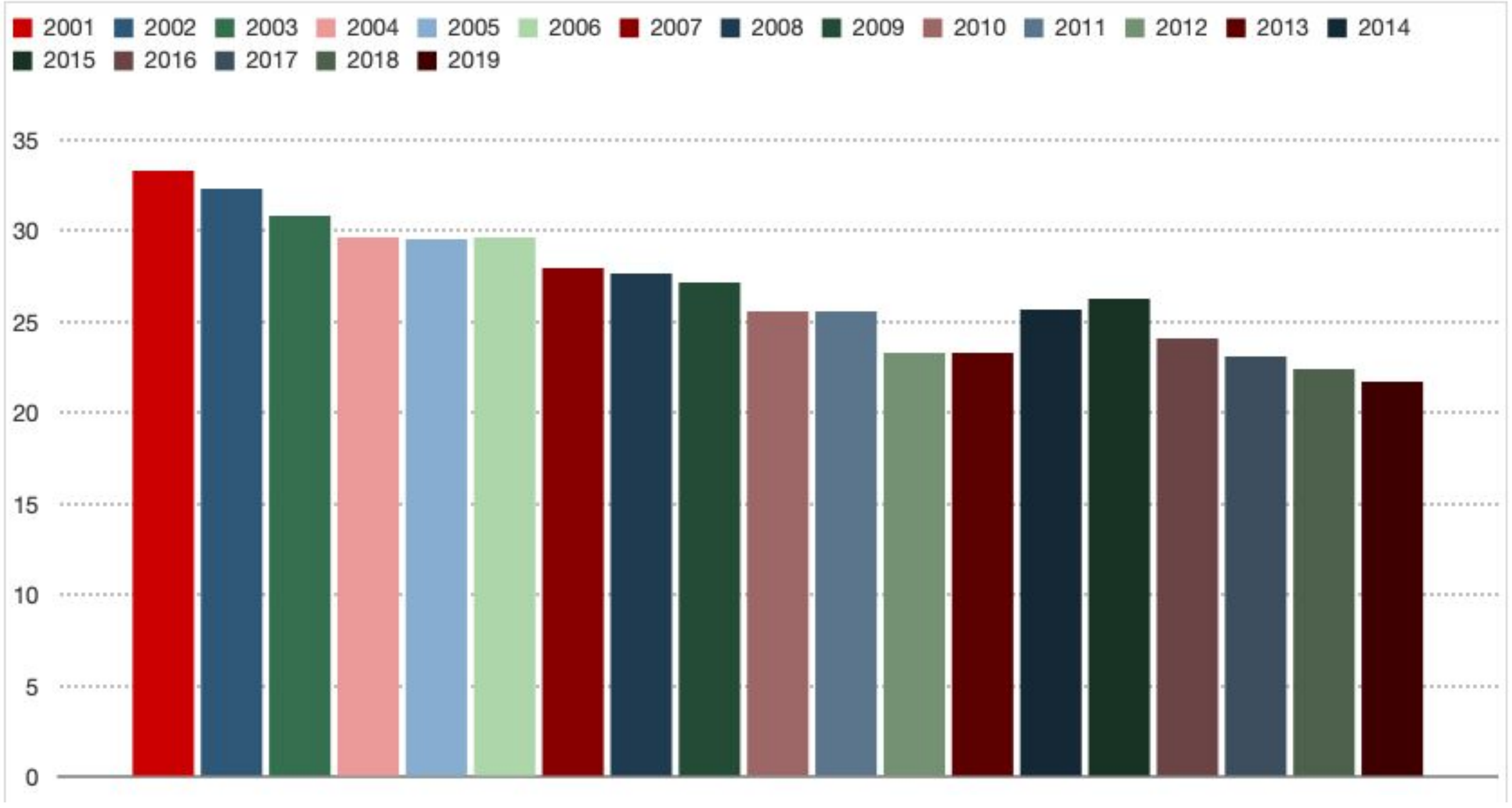
…more helpful than this?

Why is this…

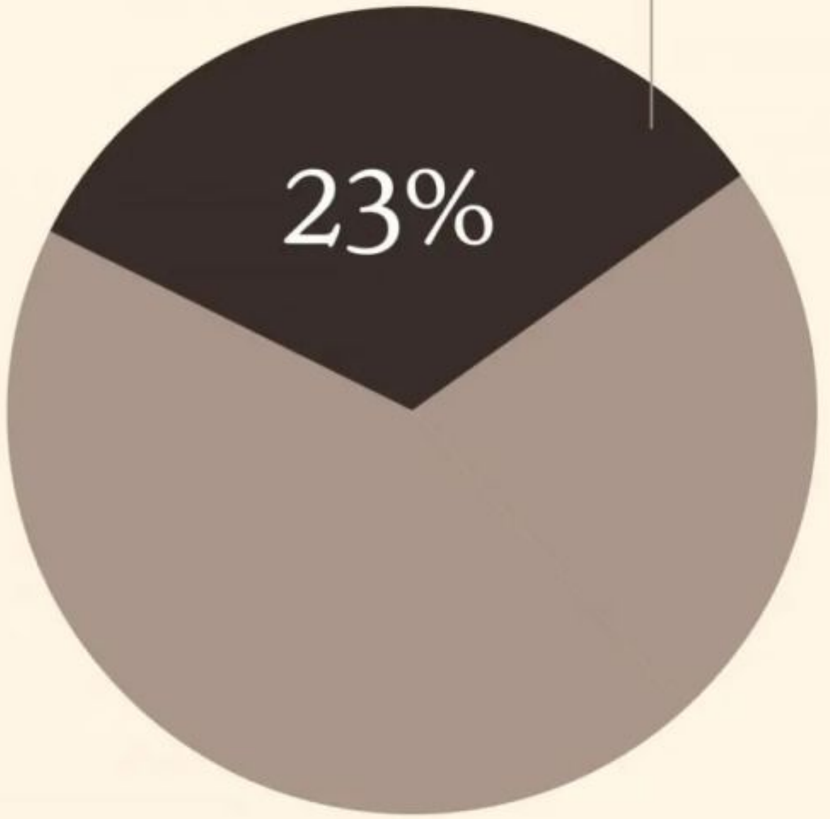


For each of the following visualizations…

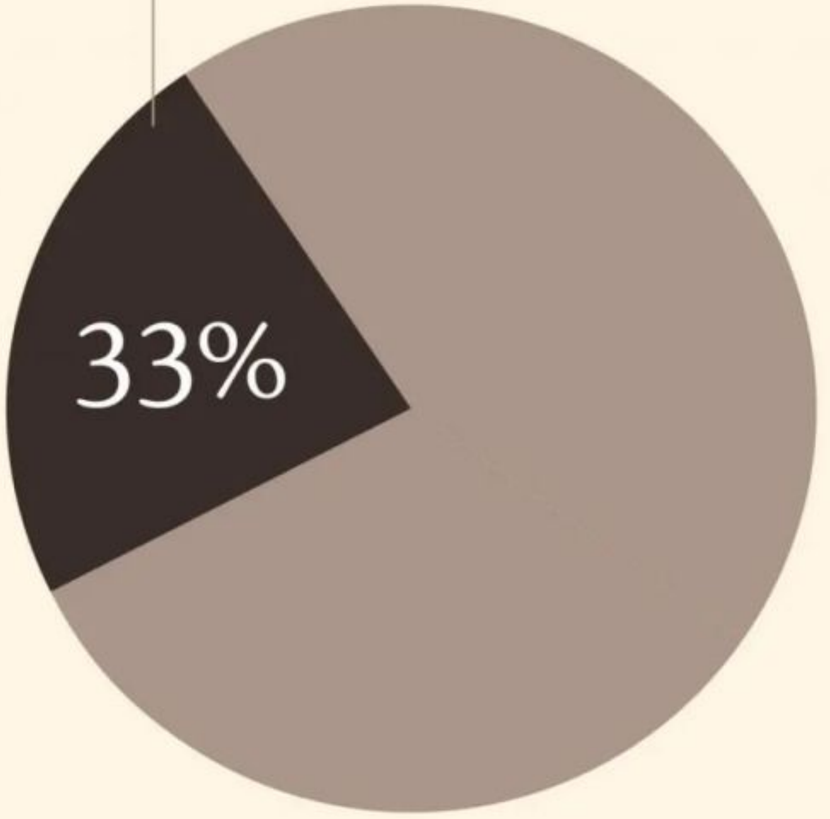
1. What is the visualization *trying* to communicate?
2. What's wrong with the visualization?
3. How would you fix it?


2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
35
30
25
20
15
10
5
0

Did not see a doctor when they needed to because of fear of being mistreated as a transgender person.



Reported having at least one negative experience related to being transgender with a health care provider.



National Center for Transgender Equality. US Transgender Survey, 2015. Available at: http://www.ustranssurvey.org

VIOLENT CRIME IS A VERY BIG PROBLEM
ADULTS



THE WIZARD OF ODDS

WHITE HOUSE PREPARES TO ADDRESS SURGE IN VIOLENT CRIME

LIVE
CNN
6:32 PM PT

"LESS EAGER TO GET THE SHOT," JEFF ZIENTS SAYS ▶ BIDEN WANTED 70

CUOMO PRIME TIME

HOW MUCH DO YOU SPEND ON GROCERIES EVERY WEEK?



22% UNDER $100

26% ABOUT $100

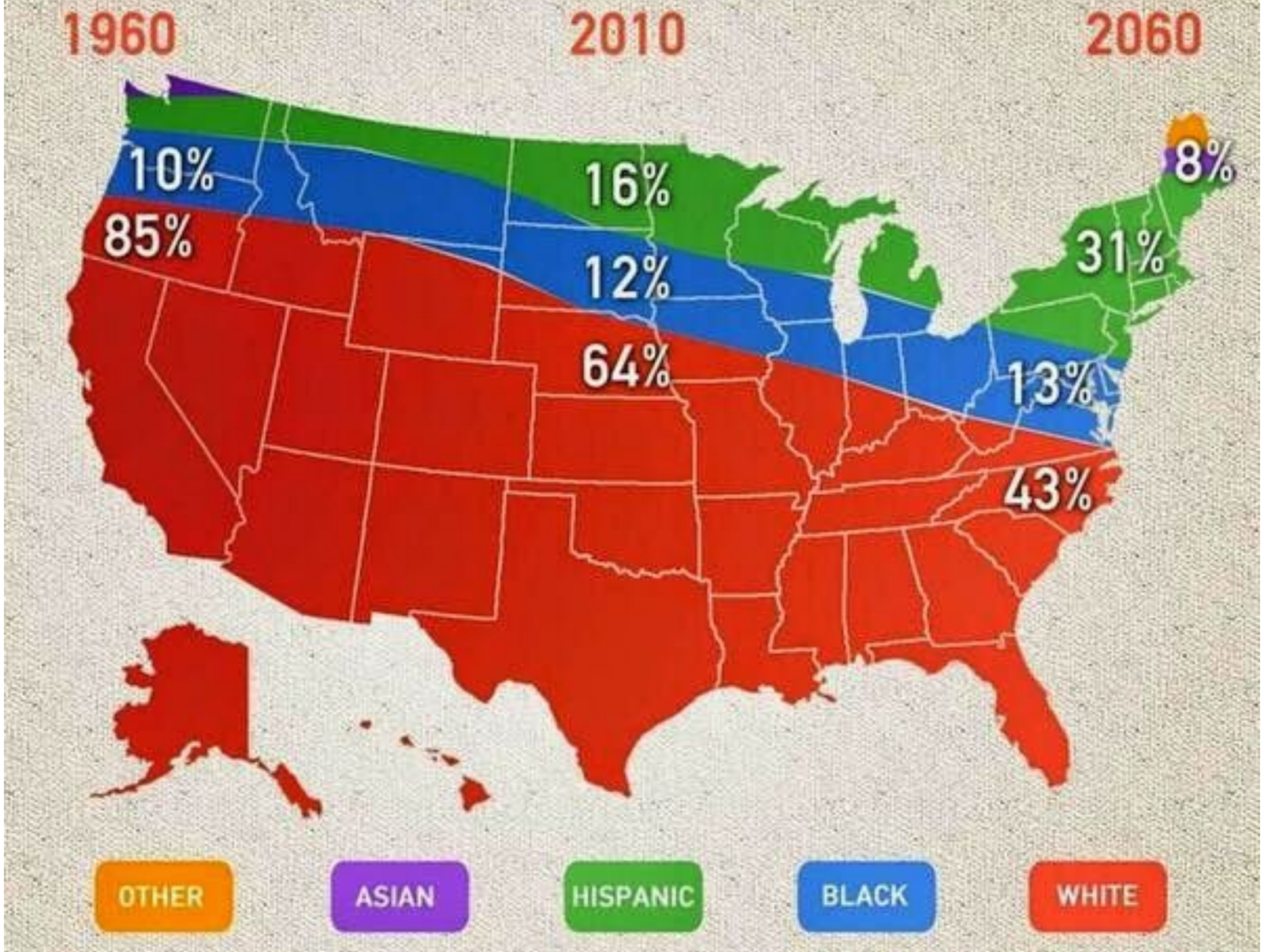
39% $100 TO $200

10% $200 TO $300

3% MORE THAN $300

CHANGING FACE OF AMERICA

Percent of total U.S. population by race and ethnicity, 1960-2060



https://informationisbeautiful.net/

https://www.reddit.com/r/dataisbeautiful/

COGS 128 Information Visualization

DSC 106 Intro to Data Visualization

Takeaways

- Data visualizations are a tool to make patterns in data more visible and/or obvious
- Good visualizations make the main message easy to see
- Visual encodings determine how easily and accurately people understand data
- Visualization choice should be driven by your data type and question
- **Next week:** Using visualizations in exploratory data analysis (EDA) to understand and describe patterns in data